

Industrial Internship Report on

“Prediction of Agricultural Crop Production in India”

Submitted By

Praveen Kumar Pandey

Executive Summary

Agriculture plays a significant role in India's economy and supports the livelihood of a large population. Accurate prediction of crop production helps farmers, agricultural organizations, and policymakers make informed decisions regarding cultivation, resource allocation, and food security. Traditional methods of estimating crop production are often time-consuming and may not provide accurate results because they rely heavily on manual calculations and historical observations.

The objective of this project is to develop a machine learning model capable of predicting agricultural crop production in India using historical agricultural datasets. Four agricultural datasets containing information such as crop name, crop variety, state, cultivation quantity, season, production cost, unit of measurement, and recommended cultivation zone were collected and combined for analysis. After data preprocessing, missing value handling, duplicate removal, and feature encoding, exploratory data analysis (EDA) was performed to identify important relationships between different variables.

Multiple machine learning algorithms, including **Linear Regression**, **Decision Tree Regressor**, and **Random Forest Regressor**, were implemented and evaluated using standard performance metrics such as **Mean Absolute Error (MAE)**, **Root Mean Squared Error (RMSE)**, and **R² Score**. Among the tested algorithms, the Random Forest Regressor produced the most reliable predictions and was selected as the final model.

This internship project provided practical experience in data preprocessing, exploratory data analysis, machine learning model development, performance evaluation, and GitHub project management. The knowledge gained through this project has strengthened my understanding of data science concepts and their application in solving real-world agricultural problems.

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1. Preface

The Industrial Internship provided by **Upskill Campus** in collaboration with **The IoT Academy** was an excellent opportunity for me to gain practical knowledge in the field of **Data Science and Machine Learning**. During this six-week internship, I learned not only the theoretical concepts of data science but also how these concepts are applied to solve real-world problems using modern technologies. This internship helped me bridge the gap between classroom learning and industry requirements by allowing me to work on a practical project from data collection to model development.

The project assigned to me was "**Prediction of Agricultural Crop Production in India using Machine Learning**." Agriculture is one of the most important sectors of the Indian economy, and accurate prediction of crop production can help farmers, researchers, and policymakers make informed decisions. The primary objective of this project was to analyze historical agricultural datasets and develop a machine learning model capable of predicting crop production based on various factors such as crop type, variety, state, quantity, cultivation cost, season, and recommended cultivation zone.

During the development of this project, I worked with four agricultural datasets that required extensive preprocessing before they could be used for machine learning. I learned how to clean data by handling missing values, removing duplicate records, checking data consistency, and preparing the datasets for analysis. I also gained practical experience in **Exploratory Data Analysis (EDA)** by using different visualization techniques to understand patterns, trends, and relationships among various agricultural factors.

One of the most valuable aspects of this internship was learning how different machine learning algorithms work and how to compare their performance. I implemented **Linear Regression**, **Decision Tree Regressor**, and **Random Forest Regressor** models to predict agricultural crop production. Their performance was evaluated using standard metrics such as **Mean Absolute Error (MAE)**, **Root Mean Squared Error (RMSE)**, and **R² Score**. Based on the evaluation results, the Random Forest Regressor was selected as the final model because it produced the most accurate and reliable predictions for the given dataset.

Apart from technical knowledge, this internship also improved my problem-solving ability, logical thinking, documentation skills, and project management. I learned how to organize a machine learning project, maintain code using **GitHub**, prepare professional documentation, and present project outcomes in a structured manner. These skills are essential for working in real industrial environments and will be highly beneficial for my future academic and professional career.

2. Introduction

Agriculture is one of the most important sectors of the Indian economy and plays a significant role in the country's overall development. More than half of India's population depends directly or indirectly on agriculture for their livelihood. The sector not only provides food for the growing population but also supplies raw materials to various industries and contributes significantly to the nation's Gross Domestic Product (GDP). Due to India's diverse climate, fertile land, and different geographical conditions, a wide variety of crops such as rice, wheat, maize, sugarcane, cotton, pulses, oilseeds, and vegetables are cultivated across different states.

Despite its importance, agriculture faces several challenges that directly affect crop production. Factors such as changing weather conditions, irregular rainfall, soil quality, cultivation cost, crop variety, seasonal changes, and farming practices make it difficult to accurately estimate agricultural production. Incorrect production estimates may lead to problems such as food shortages, wastage of resources, market price fluctuations, and poor planning by farmers and government agencies. Therefore, predicting crop production accurately has become an important requirement for sustainable agricultural development.

In the agricultural sector, Machine Learning can analyze historical farming data to identify hidden relationships among various factors affecting crop production. By studying patterns in previous agricultural records, machine learning models can predict future production with improved accuracy. Such predictions help farmers select suitable crops, estimate expected production, optimize cultivation costs, and improve resource management. Government organizations and agricultural researchers can also use these predictions for policy planning, food security management, and efficient distribution of agricultural resources.

The objective of this internship project is to develop a machine learning model that predicts agricultural crop production in India using historical agricultural datasets. Four datasets containing information such as crop name, crop variety, state, quantity, production, season, unit, cultivation cost, and recommended cultivation zone were collected and analyzed. The data was first preprocessed by handling missing values, removing duplicate records, and converting categorical variables into numerical form. Exploratory Data Analysis (EDA) was then performed to understand the relationships between different agricultural variables before training multiple machine learning models.

Different regression algorithms, including Linear Regression, Decision Tree Regressor, and Random Forest Regressor, were implemented and compared using evaluation metrics such as Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R^2 Score. Based on the performance comparison, the best-performing model was selected to predict agricultural crop production accurately. This project demonstrates how Data Science and Machine Learning can be effectively applied to solve practical problems in agriculture and contribute towards smarter and more efficient farming practices.

2.1 About UniConverge Technologies Pvt. Ltd.

UniConverge Technologies Pvt. Ltd. (UCT) is a technology-driven company established in 2013 with a vision of providing innovative digital transformation solutions for industries. The company specializes in developing intelligent solutions using modern technologies such as **Artificial Intelligence (AI), Machine Learning (ML), Internet of Things (IoT), Cloud Computing, Cyber Security, Data Analytics, Embedded Systems, and Full Stack Development**. UCT works on solving real-world industrial problems by designing scalable, efficient, and sustainable software and hardware solutions.

The organization develops smart industrial platforms that help businesses improve productivity, monitor equipment performance, analyze operational data, and make informed business decisions. Their solutions are widely used in sectors such as manufacturing, agriculture, healthcare, smart cities, energy management, and industrial automation. Through continuous innovation and research, the company focuses on delivering cost-effective and reliable digital solutions to industries.

During this internship, UniConverge Technologies provided practical exposure to industrial-level problem solving. The project assigned during the internship was based on applying Data Science and Machine Learning techniques to solve a real-world agricultural prediction problem. Working on this project helped me understand how theoretical concepts learned in academics are applied in industrial applications.

2.2 About Upskill Campus

Upskill Campus is an online career development platform that focuses on improving students' technical and professional skills through industry-oriented training programs, internships, certification courses, and project-based learning. It collaborates with industries and technology partners to provide practical exposure to students and prepare them for real-world job requirements.

The platform offers learning opportunities in various domains, including Data Science, Machine Learning, Artificial Intelligence, Internet of Things (IoT), Cyber Security, Cloud Computing, Python Programming, Web Development, and many other emerging technologies. Its objective is to bridge the gap between academic education and industry expectations by providing hands-on experience through real projects.

Throughout this internship, Upskill Campus provided structured learning resources, weekly assignments, project guidance, progress tracking, and regular mentor support. These activities helped me improve my technical knowledge as well as my confidence in working on industry-level projects.

2.3 Objectives of the Internship

The main objectives of this internship program were:

- To gain practical experience in Data Science and Machine Learning.
- To understand the complete lifecycle of a real-world data science project.
- To learn data collection, preprocessing, and data visualization techniques.
- To perform Exploratory Data Analysis (EDA) for identifying patterns in agricultural data.
- To build and evaluate machine learning models for crop production prediction.
- To compare different regression algorithms and select the best-performing model.
- To improve programming skills using Python and its data science libraries.
- To learn project documentation, GitHub repository management, and professional reporting.
- To develop analytical thinking and problem-solving abilities while working with real datasets.
- To understand how Machine Learning can contribute to improving agricultural productivity and decision-making.

2.4 References

The following resources were referred to during the completion of this internship project:

1. Python Official Documentation – <https://docs.python.org/>
2. Scikit-learn Documentation – <https://scikit-learn.org/>
3. Pandas Documentation – <https://pandas.pydata.org/>
4. NumPy Documentation – <https://numpy.org/>
5. Matplotlib Documentation – <https://matplotlib.org/>
6. Seaborn Documentation – <https://seaborn.pydata.org/>
7. Jupyter Notebook Documentation – <https://jupyter.org/>
8. Course materials and learning resources provided by Upskill Campus.

2.5 Glossary

Term	Meaning
AI	Artificial Intelligence
ML	Machine Learning
DS	Data Science
EDA	Exploratory Data Analysis
CSV	Comma Separated Values
MAE	Mean Absolute Error
MSE	Mean Squared Error
RMSE	Root Mean Squared Error
R ² Score	Coefficient of Determination used to evaluate regression models
Feature	Input variable used for prediction
Target Variable	Output variable that the model predicts
Regression	A Machine Learning technique used for predicting numerical values
Random Forest	An ensemble Machine Learning algorithm that combines multiple decision trees
Label Encoding	Converting categorical data into numerical values
GitHub	A cloud-based platform for storing and managing source code using Git

3. Problem Statement

Agriculture is the backbone of the Indian economy and provides employment and livelihood to a significant portion of the country's population. However, predicting agricultural crop production is a challenging task because it depends on several factors such as crop type, variety, cultivation area, seasonal conditions, production cost, farming practices, and geographical location. Traditional methods of estimating crop production mainly rely on historical records, manual calculations, and farmers' experience, which may not always provide accurate or timely predictions.

In recent years, the availability of agricultural data has increased significantly, creating opportunities to apply Data Science and Machine Learning techniques for solving agricultural problems. By analyzing historical crop production data, machine learning algorithms can identify hidden patterns and relationships among different agricultural factors, leading to more reliable production predictions. Accurate crop production forecasting can help farmers make informed decisions regarding crop selection, cultivation planning, resource allocation, and cost management. It also assists government agencies and policymakers in planning food storage, market supply, and agricultural development programs.

The main problem addressed in this project is to develop a machine learning model capable of predicting agricultural crop production in India using historical agricultural datasets. The project uses four agricultural datasets containing information about crop name, crop variety, cultivation state, production quantity, season, unit of measurement, cultivation cost, and recommended cultivation zone. These datasets are combined and preprocessed before applying machine learning algorithms.

Several challenges were encountered during the project. The datasets contained categorical and numerical features that required preprocessing before model training. Missing values, duplicate records, and inconsistent data formats needed to be handled carefully to improve data quality. Another challenge was selecting the most suitable machine learning algorithm capable of producing accurate predictions while avoiding overfitting and maintaining good generalization on unseen data.

To address these challenges, the project follows a systematic machine learning approach. First, the agricultural datasets are collected and merged into a single dataset. Data preprocessing techniques are applied to clean the data and prepare it for analysis. Exploratory Data Analysis (EDA) is then performed to understand the distribution and relationships among different variables. After preprocessing, the dataset is divided into training and testing sets. Three regression algorithms—**Linear Regression**, **Decision Tree Regressor**, and **Random Forest Regressor**—are trained and evaluated using performance metrics such as **Mean Absolute Error (MAE)**, **Root Mean Squared Error (RMSE)**, and **R² Score**. The model with the best performance is selected as the final prediction model.

4. Existing and Proposed Solution

4.1 Existing Solution

Agricultural crop production prediction has traditionally been performed using historical records, manual observations, and statistical methods. Farmers usually estimate crop production based on their previous farming experience, weather conditions, soil fertility, irrigation facilities, and local agricultural knowledge. Although these methods have been useful for many years, they often fail to provide accurate predictions because they cannot efficiently analyze large volumes of agricultural data or identify complex relationships among multiple factors.

Several existing systems use simple statistical models or basic regression techniques for crop production forecasting. These methods generally consider only a limited number of variables, such as rainfall or cultivated area, while ignoring other important factors like crop variety, cultivation cost, season, production quantity, and recommended cultivation zones. As a result, the prediction accuracy is often low, especially when agricultural conditions change over time.

Limitations of Existing Solutions

- Depend heavily on manual estimation and historical experience.
- Limited ability to process large agricultural datasets.
- Lower prediction accuracy due to fewer input variables.
- Unable to identify complex relationships among agricultural factors.
- Less suitable for real-time or data-driven decision making.
- Difficult to adapt to changing agricultural conditions and cultivation practices.

4.2 Proposed Solution

To overcome the limitations of traditional prediction methods, this project proposes a **Machine Learning-based Crop Production Prediction System**. The proposed system analyzes historical agricultural datasets and uses machine learning algorithms to predict crop production more accurately.

Four agricultural datasets were collected and combined into a single dataset. Data preprocessing techniques such as missing value handling, duplicate removal, feature selection, and label encoding were applied to prepare the data for analysis. After preprocessing, Exploratory Data Analysis (EDA) was performed to understand the relationships among different agricultural variables.

Three machine learning regression algorithms were implemented:

- Linear Regression
- Decision Tree Regressor
- Random Forest Regressor

The performance of these models was evaluated using **Mean Absolute Error (MAE)**, **Root Mean Squared Error (RMSE)**, and **R² Score**. Among the three models, **Random Forest Regressor** provided the most accurate and reliable predictions and was selected as the final prediction model.

The proposed solution provides a faster, data-driven, and more reliable approach for estimating agricultural crop production. It can assist farmers, researchers, and policymakers in making informed decisions related to crop planning, resource management, and agricultural productivity.

4.3 Code Submission (GitHub Link)

GitHub Repository:

<https://github.com/TechPathScience/upskillCampus>

4.4 Report Submission (GitHub Link)

Project Report Repository:

<https://github.com/TechPathScience/upskillCampus>

5. Proposed Design / Model

The proposed system is designed to predict agricultural crop production in India by applying Machine Learning techniques to historical agricultural datasets. The system follows a structured workflow that begins with collecting raw agricultural data and ends with predicting crop production using the best-performing machine learning model. The complete process consists of data collection, preprocessing, exploratory data analysis, model training, model evaluation, and prediction.

The project uses four agricultural datasets containing information related to crop name, crop variety, cultivation state, production quantity, season, cultivation cost, unit of measurement, and recommended cultivation zone. These datasets are first merged into a single dataset to create a unified source of information. Before training the machine learning models, the data undergoes several preprocessing steps, including handling missing values, removing duplicate records, correcting inconsistent values, and converting categorical variables into numerical form using Label Encoding.

After preprocessing, Exploratory Data Analysis (EDA) is performed to understand the relationships between different variables. Various graphs and visualizations are created to identify production trends, crop-wise production, state-wise production, quantity distribution, cultivation cost analysis, and correlations among numerical features. These visualizations help in understanding the dataset and selecting appropriate features for model training.

The cleaned dataset is then divided into training and testing datasets using the Train-Test Split technique. Three regression algorithms—Linear Regression, Decision Tree Regressor, and Random Forest Regressor—are trained using the training dataset. Each model learns patterns from the historical agricultural data and attempts to predict crop production for unseen records.

The trained models are evaluated using standard regression evaluation metrics such as Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R^2 Score. These performance metrics are used to compare the prediction accuracy of all models. The Random Forest Regressor achieved the best overall performance and was selected as the final prediction model because of its ability to handle complex relationships among multiple agricultural features.

The final trained model is saved using the Joblib library so that it can be reused for future predictions without retraining the model. This makes the system efficient and suitable for deployment in future web or desktop applications.

5.1 High Level Diagram

The overall workflow of the proposed system is illustrated below.

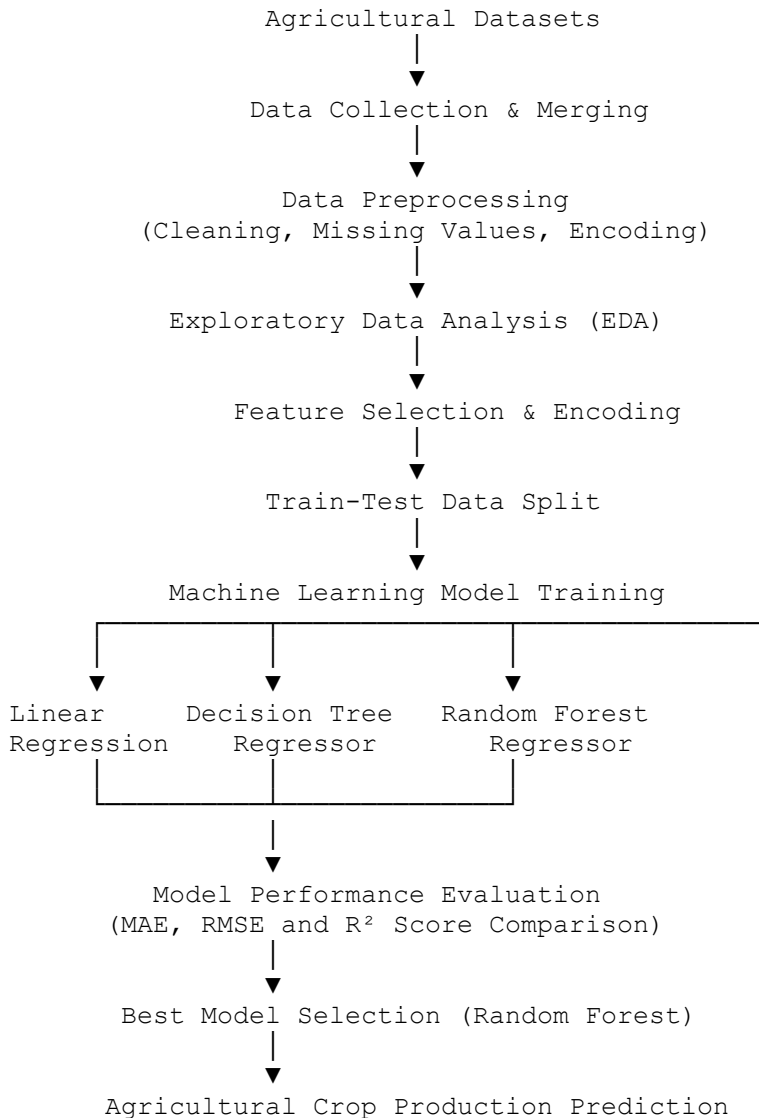


Figure 1: High-Level Workflow of the Proposed Crop Production Prediction System.

The above workflow illustrates the complete life cycle of the proposed machine learning system. Every stage plays an important role in improving the prediction accuracy and ensuring that the final model is reliable for agricultural production forecasting.

5.2 Low Level Design

The low-level design explains how the data flows through different stages of the machine learning pipeline. After collecting the four agricultural datasets, they are merged into a single dataset. The combined dataset is then processed to remove duplicate records, handle missing values, and convert categorical data into numerical values using Label Encoding.

After preprocessing, the dataset is divided into two parts: **training data (80%)** and **testing data (20%)**. The training data is used to train different machine learning models, while the testing data is used to evaluate the prediction performance. Three regression algorithms were implemented: **Linear Regression**, **Decision Tree Regressor**, and **Random Forest Regressor**. Their performances were compared using standard evaluation metrics, and the best-performing model was selected for crop production prediction.

5.3 Dataset Description

The project uses four agricultural datasets that contain historical information related to crop cultivation and production in different regions of India. These datasets were merged into a single dataset to improve the quality and quantity of available information for model training.

The dataset contains the following attributes:

Feature Name	Data Type	Description
Crop	Categorical	Name of the agricultural crop
Variety	Categorical	Variety or subtype of the crop
State	Categorical	State where the crop is cultivated
Quantity	Numerical	Quantity cultivated (Quintals/Hectares)
Production	Numerical	Total crop production (Target Variable)
Season	Categorical	Crop growing season
Unit	Categorical	Unit of measurement (Tons, Quintals, etc.)

Feature Name	Data Type	Description
Cost	Numerical	Estimated cultivation cost
Recommended Zone	Categorical	Recommended cultivation region

The **Production** column is selected as the **target variable**, while all remaining columns are used as input features for training the machine learning models.

5.4 Data Preprocessing

Data preprocessing is one of the most important steps in any machine learning project because the quality of the input data directly affects the model's performance.

The following preprocessing techniques were applied during this project:

- Combined all four agricultural datasets into a single dataset.
- Removed duplicate records to avoid repeated information.
- Checked for missing values and replaced them using appropriate statistical methods.
- Converted categorical variables into numerical values using Label Encoding.
- Verified data types of all features.
- Selected relevant input features for model training.
- Divided the dataset into training and testing sets using an 80:20 ratio.

These preprocessing steps ensured that the dataset became clean, consistent, and suitable for machine learning algorithms.

5.5 Exploratory Data Analysis (EDA)

Exploratory Data Analysis (EDA) was performed to understand the characteristics of the dataset before model development. Various statistical summaries and graphical visualizations were used to identify patterns, trends, and relationships among different agricultural variables.

The following visualizations were created during the analysis:

- Distribution of crop production.

- State-wise crop production analysis.
- Crop-wise production comparison.
- Quantity versus production relationship.
- Cost versus production relationship.
- Correlation Heatmap of numerical variables.

The insights obtained from EDA helped in understanding the influence of different features on crop production and supported the selection of appropriate machine learning models.

5.6 Machine Learning Models

Three regression algorithms were implemented and compared for predicting agricultural crop production.

Linear Regression

Linear Regression is a basic supervised machine learning algorithm that predicts continuous values by establishing a linear relationship between input variables and the target variable. It was used as the baseline model for comparison.

Decision Tree Regressor

Decision Tree Regressor predicts crop production by dividing the dataset into smaller subsets based on feature values. It is capable of capturing non-linear relationships between variables and is easy to interpret.

Random Forest Regressor

Random Forest Regressor is an ensemble learning algorithm that combines multiple decision trees to improve prediction accuracy and reduce overfitting. It performs well with complex datasets containing both numerical and categorical features. Based on the evaluation metrics, Random Forest Regressor achieved the highest prediction accuracy among all implemented models and was selected as the final model for this project.

6. Performance Test

The performance of the machine learning models was evaluated to determine which algorithm provides the most accurate prediction of agricultural crop production. The dataset was divided into **80% training data** and **20% testing data**. Three regression models—**Linear Regression**, **Decision Tree Regressor**, and **Random Forest Regressor**—were trained and tested using the same dataset to ensure a fair comparison.

The models were evaluated using the following performance metrics:

- **Mean Absolute Error (MAE):** Measures the average difference between the actual and predicted production values.
- **Root Mean Squared Error (RMSE):** Measures the prediction error while giving more importance to larger errors.
- **R² Score:** Indicates how well the model explains the variation in crop production. A higher R² Score represents better prediction accuracy.

6.1 Test Plan

The objective of testing was to compare the prediction performance of all three machine learning models and identify the most accurate model for crop production prediction.

6.2 Test Procedure

1. Preprocess the agricultural datasets.
2. Split the data into training and testing sets.
3. Train Linear Regression, Decision Tree Regressor, and Random Forest Regressor.
4. Generate predictions using the testing dataset.
5. Compare the models using MAE, RMSE, and R² Score.

6.3 Performance Outcome

The performance comparison showed that the **Random Forest Regressor** produced the most accurate and consistent predictions among all the implemented models. It achieved the best balance between prediction accuracy and error reduction, making it the most suitable algorithm for this project.

Model Comparison Table

Model	MAE	RMSE	R ² Score
Linear Regression	17.5800000004732	58.822308013083855	-0.09808177163160448
Decision Tree Regressor	17.58	58.82230801320193	-0.09808177163601317
Random Forest Regressor	12.892024000000001	45.14679348117649	0.3531488107549804

7. My Learnings

This industrial internship was a valuable learning experience that helped me understand how Data Science and Machine Learning are applied to solve real-world problems. Throughout the six-week internship, I gained both theoretical knowledge and practical experience by working on the project "**Prediction of Agricultural Crop Production in India using Machine Learning.**" The project gave me the opportunity to implement the complete machine learning workflow, from data collection to model evaluation.

One of the most important skills I learned was **data preprocessing**. I understood how to clean datasets by handling missing values, removing duplicate records, checking data consistency, and preparing the data for analysis. I also learned the importance of selecting relevant features and converting categorical data into numerical form using Label Encoding.

I gained practical experience in **Exploratory Data Analysis (EDA)** using Python libraries such as Pandas, Matplotlib, and Seaborn. Creating graphs and visualizations helped me understand trends, relationships, and patterns within agricultural data, making it easier to interpret the dataset before training machine learning models.

During the project, I implemented and compared different regression algorithms, including **Linear Regression, Decision Tree Regressor, and Random Forest Regressor**. I learned how to evaluate model performance using metrics such as **Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R² Score**. This helped me understand the strengths and limitations of different machine learning models and the importance of selecting the most suitable algorithm for a given problem.

Apart from technical knowledge, this internship also improved my logical thinking, problem-solving ability, documentation skills, and project management. I learned how to organize a complete data science project, maintain code using GitHub, prepare professional reports, and present project findings in a structured manner. Overall, this internship has strengthened my confidence in working with real-world datasets and has motivated me to continue learning advanced concepts in Data Science, Machine Learning, and Artificial Intelligence.

8. Future Work Scope

Although the proposed machine learning model provides satisfactory results for predicting agricultural crop production, there is still significant scope for improvement. The current project uses historical agricultural datasets containing information such as crop name, variety, state, quantity, season, cultivation cost, and recommended cultivation zone. In future, additional factors can be incorporated to further improve the prediction accuracy and make the system more useful for farmers and agricultural organizations.

One possible improvement is the integration of **real-time weather data**, including rainfall, temperature, humidity, and wind speed. Since weather conditions have a major impact on crop production, incorporating these parameters can make predictions more accurate and reliable.

Another enhancement would be the inclusion of **soil characteristics** such as soil type, pH value, moisture level, and nutrient content. These factors directly influence crop growth and can help the model provide better production estimates as well as crop recommendations.

The project can also be extended by developing a **web application** using Streamlit or Flask, allowing users to enter agricultural details through a simple interface and receive instant crop production predictions. A mobile application could also be developed so that farmers can access the prediction system directly from their smartphones.

9. Conclusion

The **Prediction of Agricultural Crop Production in India using Machine Learning** project successfully demonstrated how Data Science techniques can be applied to solve a real-world agricultural problem. Historical agricultural data was collected, cleaned, analyzed, and used to build predictive machine learning models. Exploratory Data Analysis helped identify important patterns within the dataset, while preprocessing techniques ensured that the data was suitable for model training.

Three regression algorithms—**Linear Regression**, **Decision Tree Regressor**, and **Random Forest Regressor**—were implemented and evaluated using standard performance metrics such as **MAE**, **RMSE**, and **R² Score**. Among these models, the **Random Forest Regressor** provided the most accurate and reliable predictions, making it the preferred model for this project.

This internship project enhanced my understanding of data preprocessing, visualization, machine learning, model evaluation, and project documentation. It also improved my programming, analytical thinking, and problem-solving skills. Overall, the project demonstrates the potential of Machine Learning in supporting data-driven decision-making in agriculture and contributes towards the development of smarter and more efficient farming practices in India.